

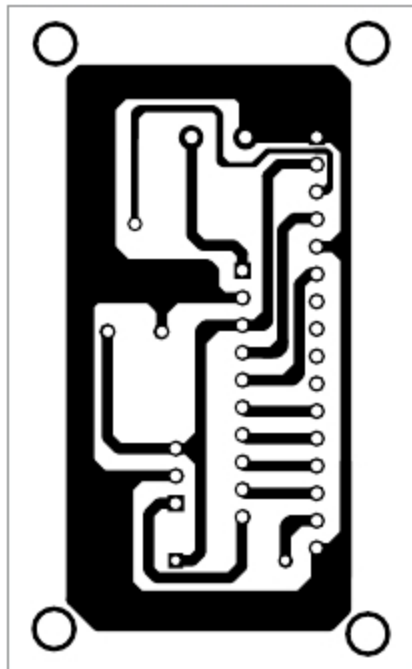


Fig. 3: Actual-size PCB layout of the noise level monitor

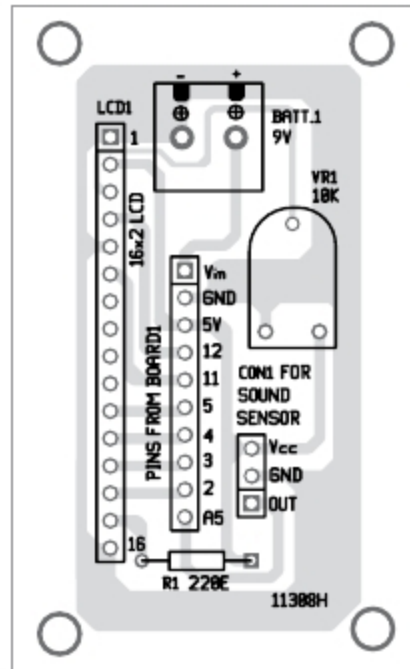


Fig. 4: Components layout for the PCB

level monitor is shown in Fig. 3 and its components layout in Fig. 4. Assemble the components on the PCB as per the circuit diagram. In this project, analogue OUT pin of the sound sensor is connected to A5 pin of Arduino, Vcc pin to 5V and GND pin to ground pin (GND) of Arduino.

Connect Arduino to a computer

using a USB cable. Open the source code from Arduino IDE, select the proper COM port and board from Tools menu. After selecting the proper COM port and board, compile and upload the source code to the board. After uploading the code, remove the USB cable and connect a 9V battery as power supply for standalone use. After 9V battery is connected to the circuit, you will see Audio Meter on LCD1. Now, speak in front of the microphone. The sound detected will be converted to corresponding analogue voltage, digital voltage and sound level in dB, and all these values will be displayed on LCD1. You can use this circuit to cap-

ture the noise around the meter.

The circuit has been tested for various situations and works fine. **EFY**



Navpreet Singh Tung is assistant professor and project coordinator in the department of electrical engineering at Bhutta Group of college, Punjab

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno board
- Sound sensor module

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 220-ohm
VR1 - 10-kilo-ohm preset

Miscellaneous:

- BATT.1 - 9V battery
LCD1 - 16x2 alphanumeric LCD
CON1 - 3-pin connector for sound sensor module
- 10-pin connector for Board1

IoT BOARD KEY FEATURES:

- Two in one** - can be used with Arduino or Raspberypi
- Five sensors onboard** - Temperature & Humidity, Gas sensor, Light sensor, Ultrasonic distance sensor and Soil moisture sensor
- Two type of displays** - LCD and LED seven segment
- Peripherals** like Relay, Buzzer, Stepper motor, Key matrix, Switches, LED's, UART, RTC are provided on board
- Students** can use the board for their project work



Customized Projects for Space, Defense, Atomic sector and other Industries.



Industrial Products: Network Time Display, Time Code Readers, Controllers and specialized test systems.



Institutional Products: Microcontroller & Microprocessor Trainers, ARM Boards, PLC Trainers, PCI cards, FPGA boards, Project boards (ARM7, ARM CORTEX, FPGA, AT89c51ED2 and ATMEGA and PIC), Communication & RF trainers and Microwave & Antenna



ADVANCED ELECTRONIC SYSTEMS

#143, 9th Main Road, Near Laggere Cross, 3rd Phase, Peenya Industrial Area, Bangalore - 560 058. Email: alsindia.net@gmail.com, sales@alsindia.net
Mob.: 98860 59662, 080-40965575, Website: www.alsindia.net